Native Candidate Probabilities from a 2B Decoder on a Single Consumer GPU

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Abstract

Applications that use language models to make decisions usually need a probability for each possible outcome. Reading answer-letter probabilities from a generative model makes them depend on the order of the options, and probabilities the model writes down must be parsed and may not form a distribution. We describe Vej, a 2B-parameter decision model that scores each developer-supplied outcome with a scalar head on a causal decoder (Qwen3.5-2B with LoRA) and normalises the scores within a question, so its output is a distribution over exactly the supplied outcomes, with no generation. In a pre-registered comparison on 14,286 decisions from dialogue domains absent from its training data and from held-out rating prompts, Vej reaches 76.60% accuracy and an NLL of 0.589. Zero-shot prompted Qwen3.5 models of 2B and 4B parameters reach at most 55.66% accuracy and at best an NLL of 1.134, and all eight pre-registered comparisons favour Vej; a 9B model in FP8, reported descriptively, reaches 62.30% and an NLL of 1.003. Three of the four prompted 2B and 4B baselines answer one task with a constant, letter-mode answers follow option position, and written probabilities are often invalid or one-hot. Vej's ECE is 0.053 without adjustment, though it is overconfident at confidences of 0.7–0.9. Training and serving fit on one consumer GPU, but Vej is not the fastest option: each candidate row repeats the state, so even with prefix caching it serves about half as many decisions per second as letter-mode prompting of the same backbone. On rubric ratings Vej is only a few points above a majority-label baseline in exact accuracy, and no generative model fine-tuned on the same data is included, so the comparison is between a model trained on decision data and zero-shot prompting, not between a scalar head and an equally trained generative head.

1 Introduction

Many uses of language models in software end in a decision among a small, known set of outcomes: which intent a customer has, whether a request has all the information an action needs, how well a response meets a rubric. Code then acts on the result. For that code, a probability per outcome is more useful than a single answer: it can defer uncertain cases, apply thresholds chosen for the cost of errors, or combine evidence.

Generative language models do not produce such probabilities directly; they are usually obtained in one of two ways: by reading the next-token probabilities of option letters after a multiple-choice prompt, or by asking the model to write a probability for each option. Both inherit the language-modelling head as the interface. Letter probabilities depend on the order and labels of the options [20, 32]. Written probabilities need decoding and parsing, may not form a distribution, and are often overconfident [29].

Vej takes a different route. Each outcome the developer supplies is rendered together with the state and the question into its own input row; a causal decoder with a scalar head scores the row; the scores of one question are normalised with a softmax. The model never generates text. The output is a distribution over exactly the offered outcomes, candidates carry no position labels, and training optimises the same likelihood that is evaluated. Vej is built on Qwen3.5-2B with LoRA adapters, was trained on 100,002 decisions, all but 3,072 of them converted from public datasets, on one consumer GPU, and is served by vLLM's classification runner on the same GPU.

This paper makes three contributions.

1. A description of native candidate scoring as a decision interface, including the three decision types (Choice, Noul and Score) and how each is turned into independent candidate rows (Section 3).
2. A pre-registered comparison (registered after Vej's own test results were known; Section 4) with prompted open models of the same and larger size (2B, 4B and, descriptively, 9B), in the two usual modes, on 14,286 decisions from dialogue domains absent from Vej's training data and from held-out rating prompts, with paired, dialogue-level confidence intervals (Sections 4 and 5).
3. Measurements of what it costs to train and serve the model on one consumer GPU, including throughput, latency and the effect of prefix caching, measured against the prompted baselines on the same machine and serving stack (Section 6).

We report negative results alongside positive ones. Vej's exact accuracy on helpfulness ratings is only a few points above always predicting the most common rating. We did not train a generative model on the same data, so the comparison is between a model trained for these decisions and zero-shot prompting of this model family, not between a native head and a generative head trained equally well. In earlier internal evaluations of complete tool-calling workflows, Vej's probabilities did not reliably prevent wrong actions; Section 7 gives the counts.

2 Background and related work

Probabilities from generative language models. Two families of methods obtain a distribution over a fixed set of answers from an instruction-tuned language model. The first reads the next-token probabilities of answer labels after a multiple-choice prompt [12]. It needs a single forward pass, but it discards the probability mass outside the label tokens, and it is sensitive to the order and symbols of the options: models prefer some label positions irrespective of content [20, 32]. The second asks the model to state its probabilities in text [23, 29]. Verbalised probabilities require decoding, parsing and validation; they are typically coarse (rounded to a few values) and are not guaranteed to form a distribution. Constrained decoding [27] can force syntactically valid output but not a valid distribution. Both methods inherit the language-modelling head and its output vocabulary as the decision interface.

Scoring candidates with a classifier. Cross-encoders score a (query, candidate) pair with one scalar and rank candidates by that score [18]. Entailment classifiers turn this into zero-shot classification by posing each label as a hypothesis [15, 31]. Reward models attach a scalar head to a decoder and train it on pairwise comparisons [19]; their pairwise loss is a two-candidate softmax cross-entropy. Vej uses the same kind of model and loss, generalised to any number of candidates, but for a different interface: the outcomes are supplied by the developer for each question, and three

decision types (choosing, assessing a statement, rating against a rubric) share one model. Unlike entailment classifiers it does not rely on a fixed three-way label space.

Probabilistic decision interfaces. TypeSafe's hosted Jev models [24] introduced an interface of three decision primitives over a developer-supplied state: choosing among named options, assessing whether a statement is true, and rating against an ordered rubric. Vej adopts the same three primitives (called Choice, Noul and Score) as its request contract but is an independent implementation: a model that runs locally and produces the probabilities from a classification head rather than from a hosted service. We do not evaluate hosted systems in this paper.

Calibration and proper scoring. We evaluate probabilities with strictly proper scoring rules, the logarithmic score and the Brier score [3, 8], and with the ranked probability score for ordinal outcomes [7]. Expected calibration error [17] and reliability diagrams are descriptive; temperature scaling [9] is the post-hoc calibration we apply. Risk–coverage curves [6] describe how error falls as low-confidence decisions are deferred.

Serving. vLLM [13] serves decoders with paged key–value caches and can reuse the computation of shared prompt prefixes; SGLang generalises prefix reuse across requests [33]. Vej's backbone is a hybrid of gated attention and Gated DeltaNet linear-attention layers [30]. For the linear-attention layers, a reusable prefix means a saved recurrent state rather than saved keys and values, so the benefit of prefix caching cannot be assumed and is measured in Section 6.

3 Vej: native candidate scoring

3.1 Decision interface

A request contains a state s (text or a JSON value) and one or more named, independent questions. Each question supplies its own outcomes:

- **Choice**: a question and a set of named options, each with a description. The output is a categorical distribution over the options.
- **Noul** (statement assessment): a statement, optional criteria, and descriptions of the true and false outcomes. The output is the probability that the statement is true.
- **Score**: a question and K ordered rubric levels, each with a description. The output is a distribution over levels; the expected level $\sum_k k\, p_k$ (zero-based) is computed outside the model.

Question identifiers organise the response but never enter the model, and questions do not condition each other. One question with its outcomes is one decision, the unit counted throughout.

3.2 Scoring

For a question q with candidate outcomes $c_1, \ldots, c_K$, Vej renders one input row per candidate,

$$x_i = \texttt{json}(\{\texttt{state} : s,\ \texttt{instruction} : \iota(q),\ \texttt{candidate} : c_i\}), \tag{1}$$

where $\iota(q)$ is the question text (for Noul, its criteria and statement) and c_i is the outcome's name and description (for Noul, the true or false description; for Score, the level description without its

position). A causal decoder h_θ reads the row and a linear head maps the final hidden state of the last token to a scalar:

$$z_i = w^\top h_\theta(x_i)_{\text{last}}, \qquad p(c_i \mid s, q) = \frac{\exp(z_i/T)}{\sum_{j=1}^{K} \exp(z_j/T)}, \tag{2}$$

with temperature $T = 1$ unless stated. The normalisation is ordinary code outside the network.

This design has four consequences that we test or rely on later.

1. **The output is always a distribution over exactly the supplied outcomes.** There is no text to parse, no probability mass outside the candidate set, and no invalid response.
2. **One forward pass per candidate and no decoding.** The cost of a decision is K row encodings, which run in parallel; each row repeats the state, which is the main cost of this design (Section 6).
3. **No answer labels.** Candidates are not tagged with letters or positions, so permuting the candidates permutes the distribution, up to numerical differences between batches.
4. **Shared prefixes.** All rows of a question begin with the same state and instruction, so a causal backbone can, in principle, reuse their computation (Section 6).

Two limitations follow equally directly. Candidates are scored independently, so the model cannot compare options inside one forward pass; and softmax assigns all probability to the supplied outcomes even when none is correct, so applications must offer explicit outcomes such as "none" or "not enough information".

3.3 Model

The model evaluated in this paper is Vej alpha2, released as Pea (the 2B model of the Vej family) and written Vej below; Vej alpha1 is an earlier encoder model used as a baseline (Section 4).

The backbone is the text part of Qwen3.5-2B [21] (revision `15852e8c`), with 1,881,825,088 parameters in 24 layers: 18 Gated DeltaNet linear-attention layers [30] and, in every fourth position, 6 gated full-attention layers (hidden size 2,048). The vision encoder and the language-modelling head are not used. We add LoRA adapters [11] of rank 16 and $\alpha = 32$ to all linear layers and a bias-free scalar head kept in FP32: 16,821,248 trainable parameters. The backbone stays frozen in BF16. Each row may contain at most 2,048 tokens; longer rows are rejected, never truncated.

3.4 Training

The model is trained with the categorical cross-entropy of the target outcome under the per-question softmax, so the training objective is the same likelihood that is evaluated. The training split has 100,002 decisions (68,719 Choice, 13,771 Noul and 17,512 Score) drawn from ten sources: MultiNLI [28], SNLI [2], WANLI [16], CLINC150 [14], BANKING77 [4], HH-RLHF preference pairs [1], HelpSteer and HelpSteer2 ratings [25, 26], Schema-Guided Dialogue (SGD) [22], and 3,072 decisions from authored workflow policies (Table 8 in the appendix). The latter are generated by the author from a fixed language bank: fictional security, agent-trace, invoice and customer-service records with balanced outcomes for four fixed policies. They are synthetic, not observations of deployed systems, and are not part of the supplementary material. Each source is converted into native questions while keeping its original human or source labels: NLI labels become relation questions, intent labels become Choice catalogues, ratings become Score rubrics, and SGD dialogue states

Table 1: Composition of the test set.

Source	Question	Type	Outcomes	Decisions
HelpSteer	Helpfulness rating	Score	5	5,000
SGD	Active intent	Choice	3–5	3,099
SGD	Required inputs known	Noul	2	3,089
SGD	Slot value	Choice	4–10	3,098
Total				14,286

become questions about the active intent, slot values and required-input completeness. Splits are grouped by source example or dialogue, so no dialogue or prompt appears in two roles.

Training used one consumer GPU, AdamW with learning rates 5×10^{-5} (LoRA) and 2.5×10^{-4} (head), weight decay 0.01, a linear schedule with 6% warm-up, effective batch size 16 decisions, gradient checkpointing and seed 43, for one epoch (6,251 updates). Its memory use and duration are reported in Section 6. We selected the checkpoint at update 6,144 by the lowest pooled validation NLL (0.4585). This run started from update 1,024 of an earlier, interrupted run on the same data with a fresh optimiser state, and earlier pilot stages on 4,096 and 32,768 decisions preceded it, so the exact training history is a chain rather than a single run. The validation split was used repeatedly for checkpoint selection, and 12 of its WANLI decisions were exposed to training in an earlier stage; it is held out from training but not untouched.

3.5 Serving

For serving, the LoRA weights are merged into the backbone and the model is saved as a sequence-classification model with one output. vLLM [13] serves it with its pooling runner, last-token pooling and an identity activation, returning one raw scalar per row; normalisation stays in client code. Because the merge and the serving kernels change floating-point arithmetic, the served model is not bit-identical to the reference; we measure the difference on the full test set (Section 5).

4 Experimental setup

4.1 Test set

The test set contains 14,286 decisions in two groups (Table 1). The first is 9,286 decisions from SGD dialogues in the Flights and Hotels services, whose domains appear in no training or validation split: which intent is active (Choice, 3–5 options), which value a slot currently holds (Choice, up to 10 options including "not specified" and "no preference"), and whether all required inputs of an intent are known (Noul). The second is 5,000 HelpSteer helpfulness ratings, recast as five-level Score questions with one description per level. Their prompts are held out, but other HelpSteer prompts are part of the training data, so this group tests new prompts rather than a new task. Inputs are long: the median candidate row in the test set has 1,181 tokens, against 127 in training.

The test set was frozen before Vej's final evaluation; the predictions used here are those of that evaluation, unchanged. The test set was later run once on a hosted system, which is outside the scope of this paper (it compares open models on one GPU). One test example was viewed while setting up that run, so we do not describe the test set as unseen; no model or prompt in this paper was selected on it. Decisions are correlated: the 9,286 SGD decisions come from 791 dialogues (typically 12 decisions each) and the 5,000 ratings from 1,464 prompts.

4.2 Baselines

All baselines answer the same 14,286 decisions with the same candidates in the same order.

- **Controls.** Uniform probabilities, and a training-prior control that gives each offered outcome a probability proportional to its add-one-smoothed frequency as the correct outcome among training questions of the same kind.
- **Vej alpha1.** An earlier Vej model: a 184M-parameter DeBERTa-v3 [10] entailment classifier [15] adapted with the same native objective on 32,548 decisions. Its rows are limited to 384 tokens; longer decisions count as failures.
- **Prompted Qwen3.5-2B**, the instruct model from which Vej's backbone is taken (revision `15852e8c`), without Vej's fine-tuning, and **prompted Qwen3.5-4B** (revision `851bf6e8`), both in BF16, each in two modes: *letter*, which reads the next-token probabilities of the option letters and renormalises over the offered letters; and *verbalised*, which asks the model for a JSON object of probabilities.
- **Prompted Qwen3.5-9B** (revision `c2022362`) with vLLM's on-the-fly FP8 weight quantisation, in the same two modes. It is the largest of the small Qwen3.5 models and fits the GPU only with FP8 weights and a 93% memory budget. This baseline was optional in the pre-registration and is reported descriptively, outside the confirmatory hypotheses.

Thinking is disabled and decoding is greedy for every prompted baseline: with thinking enabled each decision would cost thousands of generated tokens, which changes the system being compared.

Prompts. Every prompted model receives the state rendered exactly as Vej sees it, the question, and every option's name and description, labelled A, B, C, ... in the original order. Verbalised mode uses JSON schema-guided decoding so that keys and number syntax are always well formed; a response is valid only if it gives a value in $[0, 1]$ for every option and the values sum to one within 10^{-6}. Three template variants per mode were compared on a fixed sample of 300 validation decisions, and the one with the highest mean accuracy over the 2B and 4B models was frozen before any test decision was run and reused unchanged for 9B. Prompted models use an 8,192-token context; no prompt exceeded it. All templates and their validation results are in Section B.

Failures. A failure (an invalid verbalised response, an input over a model's length limit, or a runtime error) counts as wrong for accuracy and receives the uniform distribution for NLL and Brier score. We also report valid rows only, and, as a pre-registered sensitivity analysis, verbalised responses that are valid except for a sum within 0.05 of one, divided by their sum (Table 11).

4.3 Metrics and statistics

The primary metrics are exact accuracy (argmax, with ties resolved to the first candidate) and mean negative log-likelihood (NLL) of the target, with probabilities floored at 10^{-15}. Secondary metrics are the summed multiclass Brier score, 10-bin top-label expected calibration error (ECE), risk–coverage, and, for Score, the absolute error of the expected level, the ranked probability score (RPS) and within-one accuracy.

Before running any baseline we committed a pre-registration (Section E; commit `f1dcb99` in the project's repository, which is not public) that fixes the baselines, prompting protocol, metrics and two confirmatory hypotheses. Vej's test results were already known when it was written; what the pre-registration fixes is the comparison, not Vej. The hypotheses are that Vej has higher accuracy

Table 2: All systems on the 14,286 test decisions. Lower NLL, Brier (summed over outcomes) and ECE (10-bin top-label calibration error) are better. Failures count as wrong and receive a uniform distribution. "Valid" is the share of decisions with a usable distribution. Upper block: Vej and the prompted baselines (9B descriptive); lower block: the earlier model and the controls.

System	Accuracy (%)	NLL	Brier	ECE	Valid (%)
Vej alpha2	76.60	0.589	0.314	0.053	100.0
Qwen3.5-2B, letter	55.38	1.134	0.611	0.125	100.0
Qwen3.5-2B, verbalised	42.27	4.643	0.812	0.228	83.3
Qwen3.5-4B, letter	50.31	1.237	0.684	0.241	100.0
Qwen3.5-4B, verbalised	55.66	13.858	0.855	0.411	99.6
Qwen3.5-9B FP8, letter	62.30	1.003	0.528	0.161	100.0
Qwen3.5-9B FP8, verbalised	55.66	12.614	0.817	0.385	98.7
Vej alpha1 (DeBERTa)	2.34	1.375	0.723	0.271	8.5
Training prior	45.87	2.197	0.849	0.286	100.0
Uniform	35.71	1.380	0.724	0.081	100.0

Table 3: Pre-registered paired comparisons (Vej minus baseline). Positive accuracy and negative NLL differences favour Vej. Intervals resample whole dialogues and prompts; the 99.375% interval (Bonferroni over eight comparisons) decides the verdict. NLL in nats per decision. The 9B model is not confirmatory and is reported in the text.

Baseline	Metric	Vej − baseline	95% CI	99.375% CI	Verdict
Qwen3.5-2B, letter	Accuracy (pp)	+21.22	[+20.15, +22.26]	[+19.75, +22.67]	supported
	NLL	−0.544	[−0.568, −0.520]	[−0.576, −0.512]	supported
Qwen3.5-2B, verbalised	Accuracy (pp)	+34.33	[+33.17, +35.46]	[+32.71, +35.89]	supported
	NLL	−4.054	[−4.274, −3.836]	[−4.372, −3.748]	supported
Qwen3.5-4B, letter	Accuracy (pp)	+26.29	[+25.21, +27.31]	[+24.81, +27.72]	supported
	NLL	−0.647	[−0.671, −0.623]	[−0.680, −0.614]	supported
Qwen3.5-4B, verbalised	Accuracy (pp)	+20.94	[+19.96, +21.89]	[+19.59, +22.28]	supported
	NLL	−13.268	[−13.639, −12.908]	[−13.770, −12.768]	supported

(H1) and lower NLL (H2) than each prompted 2B and 4B baseline. Differences are paired by decision. Because decisions from one dialogue or prompt are correlated, confidence intervals come from a percentile bootstrap [5] that resamples whole dialogues and prompts (10,000 replicates). With eight confirmatory comparisons, a hypothesis counts as supported for a baseline when the Bonferroni-adjusted 99.375% interval excludes zero in Vej's favour, refuted when it excludes zero against Vej, and inconclusive otherwise.

5 Results

5.1 Main comparison

Table 2 lists every system on all 14,286 decisions. Vej reaches 76.60% accuracy with an NLL of 0.589. Among the prompted 2B and 4B baselines, the highest accuracy is 55.66% (Qwen3.5-4B in verbalised mode) and the lowest NLL is 1.134 (Qwen3.5-2B in letter mode). Both pre-registered hypotheses are supported against every prompted 2B and 4B baseline (Table 3): every Bonferroni-adjusted interval excludes zero in Vej's favour. The smallest accuracy gap is 20.94 percentage points and the smallest NLL gap 0.544 nats per decision.

Three failure modes of the prompted baselines account for much of the gap; a fourth observation

Table 4: Accuracy and NLL by decision type (6,197 Choice, 3,089 Noul and 5,000 Score decisions). Failures are scored as in Table 2.

	Accuracy (%)			NLL		
System	Choice	Noul	Score	Choice	Noul	Score
Vej alpha2	89.2	91.7	51.6	0.371	0.249	1.071
Qwen3.5-2B, letter	63.8	49.4	48.6	1.073	0.829	1.397
Qwen3.5-2B, verbalised	43.0	50.6	36.3	8.192	1.726	2.047
Qwen3.5-4B, letter	57.3	50.1	41.7	1.193	0.962	1.460
Qwen3.5-4B, verbalised	65.0	79.5	29.3	11.553	7.067	20.909
Qwen3.5-9B FP8, letter	74.2	73.6	40.6	0.649	0.542	1.727
Qwen3.5-9B FP8, verbalised	66.8	71.3	32.2	9.286	9.660	18.565
Vej alpha1 (DeBERTa)	0.0	0.0	6.7	1.538	0.693	1.594
Training prior	41.4	50.6	48.4	3.639	0.712	1.328
Uniform	55.3	50.6	2.3	1.538	0.693	1.609

concerns model size.

Three of the four 2B and 4B prompted systems collapse on Noul. The 3,089 Noul decisions ask whether every input that an intent requires is already known. This requires reading the intent's required slots from the schema and checking each against the dialogue. Three of the four prompted systems answer it with a constant, so their Noul accuracy in Table 4 equals the base rate of the answer they always give: Qwen3.5-2B in letter mode answers "false" for every decision, Qwen3.5-4B in letter mode for almost every decision, and Qwen3.5-2B in verbalised mode "true" for every decision, so all three are at chance. Qwen3.5-4B in verbalised mode is the exception, at 79.5%. Vej, trained on the same kind of question in other SGD services, answers 91.7% correctly.

Letter-mode Choice follows position. In this test set the first-listed option is the correct one in 55.3% of Choice decisions, mostly because "not specified" (`UNSPECIFIED`) is the first option of every slot-value question and is often correct. Uniform probabilities with ties resolved to the first option therefore score 55.3% on Choice, and Qwen3.5-2B in letter mode, at 63.8%, is only modestly above that. Reversing the option order on the fixed, stratified 1,000-decision sample also used for the serving benchmark (Section 6) changes Qwen3.5-2B's letter-mode accuracy by 7.7 percentage points, and only 46.4% of its answers stay the same; for Qwen3.5-4B, 71.5% stay the same (Table 10). Vej's candidates carry no position labels. In a post-hoc check on the 433 Choice decisions of that sample, served through vLLM, reversing the option order leaves Vej's answer unchanged in 99.8% of decisions, against 63.7% and 74.1% for the two letter-mode baselines on the same decisions.

Verbalised probabilities are invalid or overconfident. Even with schema-guided decoding, Qwen3.5-2B in verbalised mode returns a usable distribution for only 83.3% of decisions; every failure is a set of probabilities that does not sum to one. Qwen3.5-4B is almost always valid (99.6%) but almost always certain: it assigns probability 1 to one outcome in 88.8% of decisions, and puts zero probability on the correct outcome in 40.0% of its valid answers (12.1% for the 2B model). A proper scoring rule punishes this heavily: the NLL of the two verbalised baselines is 4.643 and 13.858. The probabilities are effectively a single answer written as numbers. Normalising near-valid answers (the pre-registered sensitivity analysis) does not change the conclusion (43.9% and 55.7% accuracy).

Table 5: Score decisions (5,000 five-level helpfulness ratings). MAE is the absolute error of the expected level; RPS is the ranked probability score.

System	Exact (%)	Within one (%)	MAE (levels)	RPS
Vej alpha2	51.6	95.8	0.567	0.373
Qwen3.5-2B, letter	48.6	91.5	0.651	0.480
Qwen3.5-2B, verbalised	36.3	73.9	0.804	0.558
Qwen3.5-4B, letter	41.7	85.0	0.674	0.497
Qwen3.5-4B, verbalised	29.3	75.2	1.031	0.997
Qwen3.5-9B FP8, letter	40.6	84.3	0.716	0.565
Qwen3.5-9B FP8, verbalised	32.2	78.3	0.960	0.900
Vej alpha1 (DeBERTa)	6.7	17.7	0.943	0.662
Training prior	48.4	91.3	0.691	0.494
Uniform	2.3	8.7	1.009	0.694

Scale helps, but does not close the gap. Going from 2B to 4B raises verbalised-mode accuracy from 42.27% to 55.66%, mainly through Noul, but lowers letter-mode accuracy from 55.38% to 50.31%: the 4B model's Choice answers are less concentrated on the first position, which removes part of the position advantage, and its Score answers are worse. The optional Qwen3.5-9B baseline, run with FP8 weights and a larger memory budget (deviation 2 in Section E), is the strongest prompted system: in letter mode it reaches 62.30% accuracy and an NLL of 1.003, and it no longer collapses on Noul (73.6%). Vej remains 14.30 percentage points more accurate (95% interval 13.4–15.1) with an NLL lower by 0.414. These 9B comparisons are descriptive; they were not among the confirmatory hypotheses. In verbalised mode the 9B model shows the same overconfidence as the 4B model, putting probability 1 on one outcome in 73.2% of decisions; it happens to get exactly as many decisions right as the 4B model (55.66%), although they are right on different decisions.

Input length. Vej's accuracy does not fall with input length: it is 78.5%, 71.7% and 79.7% on the shortest, middle and longest third of decisions (Figure 2). Results for each service and question kind are in Table 12.

Vej alpha1. The earlier 184M-parameter encoder model can process only 8.5% of the test decisions within its 384-token limit; on those it is right 27.7% of the time. The 2,048-token context of the 2B decoder is necessary for these inputs.

5.2 Score

Score is Vej's weakest decision type. Exact accuracy is 51.6%, against 48.4% for the training prior, which almost always predicts the most common rating (Table 5). The distributional metrics show a clearer difference: Vej's expected-level error is 0.567 levels against 0.691, its ranked probability score is 0.373 against 0.494, and 95.8% of its answers are within one level. Helpfulness ratings are subjective; exact agreement with the source-provided human rating is a hard target for any model.

5.3 Calibration

Vej's pooled calibration error is low without any adjustment: its ECE is 0.053. The pooled figure hides local overconfidence: decisions with a top probability between 0.8 and 0.9 (358 of them) have a mean confidence of 85.2% but are right only 59.2% of the time, and the 0.7–0.8 range shows

Table 6: Temperature scaling. Vej's temperature was fitted before this study on 2,026 decisions from the validation split, which had also been used for checkpoint selection; baseline temperatures are fitted on their 300-decision tuning sample. The protocol applies temperature scaling only to systems that expose logits, so verbalised baselines are not scaled.

System	T	NLL (T=1)	NLL (T)	ECE (T=1)	ECE (T)
Vej alpha2	1.33	0.589	0.557	0.053	0.038
Qwen3.5-2B, letter	1.41	1.134	1.078	0.125	0.048
Qwen3.5-4B, letter	1.68	1.237	1.056	0.241	0.129

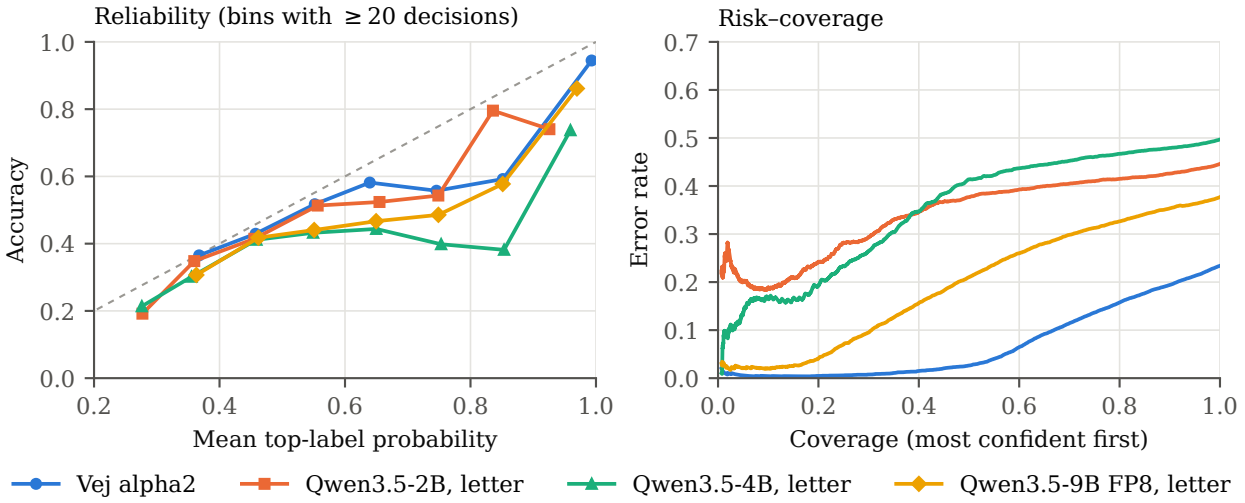


Figure 1: Calibration of Vej and the letter-mode baselines. Verbalised baselines are omitted: the 4B and 9B models put probability 1 on one outcome in most decisions, and the 2B model's output is invalid for 16.7% of decisions. Left: reliability diagram, accuracy against mean top-label probability for 10-bin intervals holding at least 20 decisions; the dashed line is perfect calibration. Right: error rate when only the most confident decisions are answered, at increasing coverage.

the same pattern (74.6% against 55.7%). These are descriptive results on this test mixture, not a calibration guarantee.

Applying the temperature fitted earlier on 2,026 validation decisions ($T = 1.33$) lowers its NLL to 0.557 and its ECE to 0.038 (Table 6). Temperature scaling also helps the letter-mode baselines, but they remain far behind on NLL, because temperature cannot fix a wrong ranking. Figure 1 shows reliability and risk–coverage curves: when Vej's top probability is at least 0.9, which is the case for 58.5% of decisions, it is wrong 5.5% of the time.

5.4 Served model

Serving Vej through vLLM changes the arithmetic: the merged weights, kernels and batching differ from the reference implementation. On all 14,286 decisions the served model gives the same top answer in 99.66% of cases, with a largest probability difference of 0.129. Its accuracy is 76.61% and its NLL 0.590, against 76.60% and 0.589 for the reference. Aggregate quality on this test is therefore similar, although individual decisions can differ, and the speed measurements in Section 6 apply to a served system whose quality was measured on the full test.

6 Running on one consumer GPU

6.1 Training

Vej's training, its evaluation and every GPU baseline in this paper ran on one consumer GPU (an earlier checkpoint and the temperature fit ran on CPUs). The selected training run peaked at 6.42 GiB (6,570 MiB) of allocated and 8.55 GiB of reserved memory with a frozen BF16 backbone, rank-16 LoRA, gradient checkpointing and one decision (all of its candidate rows) per micro-batch. Its training phase, including seven full validation passes, took 103.5 hours. This excludes the first 1,024 updates of the interrupted run and the pilot stages (Section 3). The checkpoint is a 3.6 GiB file.

6.2 Serving benchmark

We measured every system on the same GPU, with the same vLLM container image, GPU memory budget (85%) and eager execution, on a fixed, stratified sample of 1,000 test decisions. Vej is served by vLLM's pooling runner: a decision is one request containing its candidate rows, and the answer is one scalar per row. Prompted baselines are served by the generation runner with the prompts used for accuracy: letter mode generates one token and reads its log-probabilities; verbalised mode generates a JSON object with guided decoding. Prefix caching is disabled for the baselines and measured both off and on for Vej: a prompted decision is a single prompt, so it has no within-decision prefix to share, whereas Vej's candidate rows of one decision share their state and question. Every measurement starts a fresh server, so no decision is ever replayed from a warm cache: latency is measured at concurrency one on decisions 21–1,000 after 20 warm-up decisions, and throughput by releasing decisions 2–1,000 together after one first answer. Each phase was repeated three times. We also report the time from container start to the first answer (Table 7).

Qwen3.5-9B is not benchmarked: it needs a 93% memory budget (deviation 2, Section E).

Table 7: Serving on one consumer GPU (vLLM, eager mode) on a 1,000-decision test sample. Throughput: 999 decisions released together, mean and range over three fresh servers. Latency: 980 decisions at concurrency one, mean of three fresh servers. Cold start: container start to first answer, mean of all six servers. Tokens/dec.: prompt tokens per decision; Hits: vLLM's final logged prefix-cache hit rate (recent-request window), averaged over the three runs.

System	Decisions/s (range)	p50 (ms)	p95 (ms)	Tokens/dec.	Hits (%)	Cold (s)
Vej, prefix caching off	4.21 (4.20–4.23)	215	497	4,859	–	81
Vej, prefix caching on	9.38 (9.37–9.38)	102	191	4,859	58	81
Qwen3.5-2B, letter	20.88 (20.83–20.92)	57	81	1,193	–	103
Qwen3.5-2B, verbalised	15.68 (15.61–15.73)	503	816	1,332	–	104
Qwen3.5-4B, letter	8.88 (8.87–8.89)	123	182	1,193	–	104
Qwen3.5-4B, verbalised	6.04 (6.03–6.05)	947	1,290	1,332	–	105

Vej is not the fastest system. Each of Vej's candidate rows repeats the state, so a Vej decision in this sample reads 4,859 prompt tokens on average, against 1,193 for a letter-mode prompt that lists all options once. The 2B systems process prompt tokens at a similar rate, so their throughput largely follows the token count. Prefix caching lets the rows of a decision share the computation of their common state and question (vLLM reports a prefix-cache hit rate of about 58% at the end of each run, over its recent-request window); caching raises its throughput from 4.21 to 9.38 decisions per second and lowers its median latency from 215 to 102 ms. Qwen3.5-2B in letter mode remains

faster, at 20.88 decisions per second and 57 ms, with the much lower accuracy reported in Section 5. Against verbalised prompting, which must generate its answer, Vej's median latency is lower (102 ms against 503 and 947 ms), but batched decoding of the short JSON answers is efficient, so the 2B verbalised baseline still has the higher throughput (15.68 decisions per second). Vej with prefix caching has slightly higher throughput than Qwen3.5-4B in letter mode (8.88 decisions per second).

Caching does not change Vej's decisions in aggregate. Rerunning all 14,286 test decisions through vLLM with prefix caching enabled gives the same top answer as the uncached server in 99.80% of decisions, an accuracy of 76.61% and an NLL of 0.591 (uncached: 76.61% and 0.590); the largest change in a single probability is 0.093, from different kernel batching. Rows of different decisions share cache blocks only when their token prefixes are identical, so caching does not let one question condition another.

What this means. On one consumer GPU, every configuration serves a few to tens of decisions per second, with median latencies of 57–215 ms for Vej and letter-mode prompting and 503–947 ms for verbalised prompting. Vej has about half the throughput of letter-mode prompting of the same backbone; its advantage is the quality of the decisions, not their price. The repetition of the state across candidate rows is the main remaining cost; sharing it more completely, for example by caching the state once per request rather than relying on block-level prefix matching, is a direct way to reduce it.

7 Discussion and limitations

What the results show. On decisions from unseen dialogue domains and held-out rating prompts, a 2B decoder trained to score candidates natively is far more accurate, has much lower NLL and lower calibration error (at $T = 1$; after temperature scaling the 2B letter-mode ECE is 0.048 against Vej's 0.038) than the same decoder, or a larger one up to 9B parameters, prompted for probabilities. The failure modes of the prompted baselines are not subtle: constant answers on a task that requires reading a schema, answers that follow the position of the options, and written probabilities that are invalid or put zero mass on the correct outcome. Native scoring removes the interface problems by construction. How much of the remaining gap is due to training on decision data, to the scalar head, or to the input format cannot be separated without the matched baseline described below; the result is a property of the complete trained system.

What they do not show.

- **No matched generative baseline.** The prompted baselines are zero-shot. A generative model fine-tuned on the same 100,002 decisions and scored by answer-letter probabilities would separate the effect of the native head from the effect of training. We plan this comparison for a later version.
- **One test set.** The test covers two sources and four kinds of question in English. It is a transfer test, not a standard public benchmark, and the test set has also been used once to evaluate a hosted system.
- **Score.** Vej's exact accuracy on five-level helpfulness ratings is 51.6%, only modestly above the 48.4% of a majority-label prior; its advantage is in the shape of the distribution rather than in exact agreement.
- **Tool use.** In an earlier internal evaluation of 48 deliberately hard tool-calling cases (not released, no baseline), workflows driven by Vej ended correctly in 30; of the 31 cases in which

Vej's probabilities marked a call ready to execute, 9 were wrong calls. Probabilities from Vej are inputs to an application's policy, not permission to act.

- **Serving cost.** Because every candidate row repeats the state, Vej reads about four times as many tokens per decision as a letter-mode prompt; with prefix caching it is still about half as fast as the 2B letter-mode baseline. Its advantage is quality, not speed.
- **Correlated, imperfect labels.** Decisions from one dialogue or prompt are correlated, which our intervals account for, and source labels were used as published, including subjective ratings.
- **Exposure.** The backbone's pretraining data are unknown, so overlap with the public sources cannot be excluded.

Availability. The supplementary material distributed with this paper (as ancillary files) contains the identifier and SHA-256 hash of every test decision, the pinned revisions of its two sources, the per-decision predictions of Vej, every baseline and both controls, the order-reversal and cached-serving records, the template-selection scores, the sample identifiers and every benchmark run; the prompt templates are in Section B. The evaluation and analysis code depends on Vej's source and is, like the test file, Vej's weights and its training code, available from the author on request; a portable script that rebuilds the test set from the public sources is not yet available.

8 Conclusion

A decision model does not need to generate its answer. Scoring each supplied outcome with a scalar head on a small decoder gives valid probabilities that do not depend on option order by construction, without decoding. Trained mostly on public decision data, it outperforms zero-shot prompting of Qwen3.5 models at 2B and 4B on a pre-registered transfer test, and the 9B model in a descriptive comparison, with lower calibration error. Training and serving fit on one consumer GPU; serving throughput is about half that of letter-mode prompting of the same backbone. Whether the gain comes from the scalar head or from training on decision data remains open.

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Competing interests. The author develops Vej as a commercial product.

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A Training data

Table 8: Decisions per source and split. Test sources are SGD Flights/Hotels dialogues and HelpSteer ratings; SGD validation uses other service domains.

Source	Train	Validation	Test
MultiNLI	30,000	9,000	0
Schema-Guided Dialogue	16,242	8,944	9,286
HH-RLHF	15,000	2,500	0
HelpSteer2	9,000	800	0
HelpSteer	8,000	4,000	5,000
CLINC150	7,424	1,024	0
WANLI	6,144	1,024	0
BANKING77	3,072	384	0
Authored workflow policies	3,072	384	0
SNLI	2,048	512	0
Total	100,002	28,572	14,286

Licences: MultiNLI (OANC terms for the non-fiction genres used), SNLI and SGD (CC BY-SA 4.0), HelpSteer, HelpSteer2, WANLI and BANKING77 (CC BY 4.0), CLINC150 (CC BY 3.0), HH-RLHF (MIT). Source labels are used as published and were not re-adjudicated; preference and helpfulness labels are subjective, and SGD intent and completeness labels are not labels of permission to act.

B Prompt templates

All prompts use the model's chat template with thinking disabled and a generation prompt. `{state}` is the state as JSON with sorted keys, exactly as Vej receives it; `{question}` is the question text (for Noul: the criteria, a newline, and `Statement:` followed by the statement); each option line is a capital letter, a colon and the candidate text Vej scores (for Choice, the option name, a colon and its description).

Layouts. v1 and v3 use the order state, question, options; v2 uses question, options, state:

```
State:
{state}

Question:
{question}

Options:
A: {candidate 1}
B: {candidate 2}
...
```

v3 adds the system message `Evaluate the supplied decision carefully.`

Instructions. The user message ends with one of two instructions. Letter mode:

```
Choose the correct option. Reply with only its capital letter.
```

Verbalised mode, followed by the JSON schema used for guided decoding (every offered letter required, numbers between 0 and 1, no other keys):

```
Assess the probability of each option. Reply only with a JSON object mapping every option letter to
↪  a probability in [0, 1]. Probabilities must sum to 1.
JSON schema:
{schema}
```

Selection. Table 9 shows validation accuracy on the 300-decision tuning sample. The selected variants (letter v3, verbalised v2) were frozen, together with a hash of the evaluation code, before any test decision was run.

Table 9: Template variants on the 300-decision validation tuning sample.

Mode	Variant	2B (%)	4B (%)	Mean (%)
Letter	v1	53.3	60.0	56.7
Letter	v2	49.7	51.0	50.3
Letter	v3 (selected)	54.7	61.0	57.8
Verbalised	v1	36.0	59.0	47.5
Verbalised	v2 (selected)	43.7	63.0	53.3
Verbalised	v3	35.0	60.3	47.7

C Additional results

Table 10: Position bias: letter-mode accuracy on the 1,000-decision subset with the original and the reversed option order, and the share of decisions where the chosen outcome is the same. Lower block: post-hoc check restricted to the Choice decisions of the sample; Vej is served through vLLM.

System	Original (%)	Reversed (%)	Change (pp)	Same answer (%)
Qwen3.5-2B, letter	56.5	48.8	−7.7	46.4
Qwen3.5-4B, letter	50.2	53.1	+2.9	71.5
Vej alpha2 (Choice, 433)	90.8	90.5	−0.2	99.8
Qwen3.5-2B, letter (Choice, 433)	67.4	59.4	−8.1	63.7
Qwen3.5-4B, letter (Choice, 433)	57.7	68.4	+10.6	74.1

Table 11: Systems with failures: results on their valid rows only, and the pre-registered sensitivity analysis in which verbalised maps summing to within 0.05 of one (compared exactly in decimal) are divided by their sum.

	Valid rows only				Sensitivity		
System	Decisions	Accuracy (%)	NLL	Brier	Valid (%)	Accuracy (%)	NLL
Qwen3.5-2B, verbalised	11,896	50.8	5.223	0.810	86.1	43.9	4.633
Qwen3.5-4B, verbalised	14,225	55.9	13.908	0.855	99.7	55.7	13.868
Qwen3.5-9B FP8, verbalised	14,098	56.4	12.759	0.817	99.2	56.1	12.627
Vej alpha1 (DeBERTa)	1,209	27.7	1.544	0.789	–	–	–

Table 12: Accuracy (%) by SGD service (the dataset's service identifiers) and question kind, and for HelpSteer ratings. "verb.": verbalised mode.

Family	Decisions	Vej	2B letter	2B verb.	4B letter	4B verb.	9B FP8 letter	9B FP8 verb.
HelpSteer, helpfulness rating	5,000	51.6	48.6	36.3	41.7	29.3	40.6	32.2
Flights 1, slot value	1,765	94.7	71.1	50.7	73.5	79.1	85.4	81.8
Flights 1, active intent	1,771	82.0	53.4	28.8	38.7	56.6	64.4	49.4
Flights 1, required inputs	1,746	90.3	55.0	45.0	55.0	81.8	82.0	73.2
Flights 2, slot value	259	88.8	73.7	49.0	75.7	74.9	88.8	82.6
Flights 2, active intent	261	87.0	64.8	44.8	41.0	67.4	65.5	59.8
Flights 2, required inputs	258	95.0	43.4	56.6	43.8	77.1	74.0	64.0
Hotels 1, slot value	340	90.3	74.4	34.1	76.2	74.4	83.2	81.2
Hotels 1, active intent	342	84.8	64.3	61.7	38.0	45.0	65.5	49.4
Hotels 1, required inputs	343	93.3	44.3	55.7	44.6	71.7	58.0	66.8
Hotels 2, slot value	375	98.9	62.4	26.7	93.3	71.2	93.6	91.5
Hotels 2, active intent	369	85.6	59.6	52.3	26.8	48.0	50.1	42.3
Hotels 2, required inputs	384	94.8	35.4	64.6	35.9	82.8	62.5	75.8
Hotels 3, slot value	359	96.9	66.9	52.1	88.3	73.3	86.1	86.1
Hotels 3, active intent	356	88.8	64.0	57.9	30.9	41.6	55.1	55.3
Hotels 3, required inputs	358	91.6	46.1	53.9	51.7	73.7	58.9	67.3

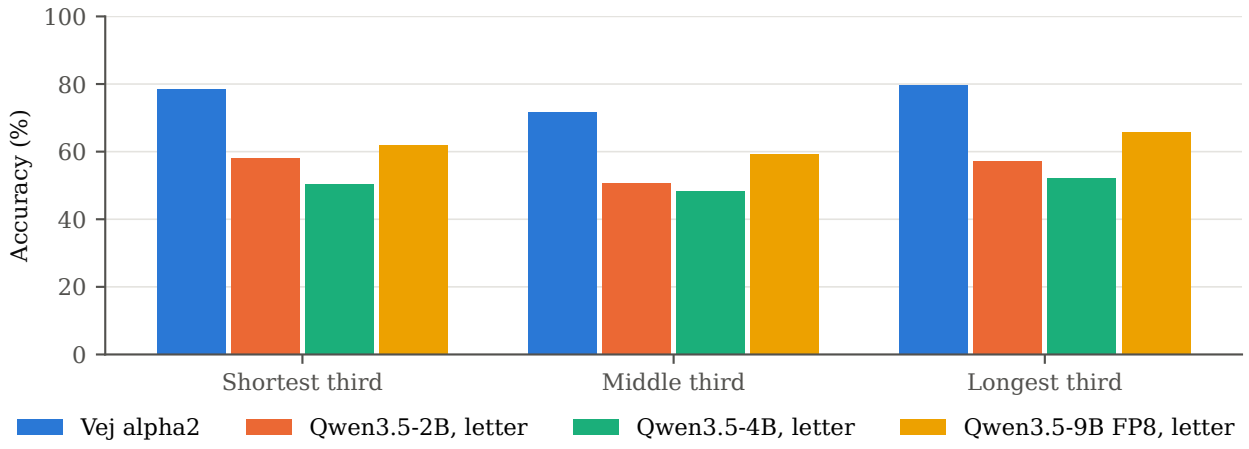


Figure 2: Accuracy by input-length tertile (rendered size of the decision). Tertiles mix decision types in different proportions, so differences between tertiles are not purely an effect of length. Verbalised baselines are not shown.

D Further attempts to improve Noul

After Vej alpha2 was frozen we ran a series of development experiments aimed at improving Noul (statement assessment), including additional sources, contrastive pairs, isolated Noul adapters and replay of earlier data. None produced a checkpoint that beat the development Noul accuracy of Vej alpha2 (92.92% on 4,182 validation decisions) while preserving its Choice and Score results, so none is evaluated here.

E Pre-registration

The pre-registration was committed to the repository before any baseline was run. Its full text, reproduced verbatim from the repository's Markdown file including the recorded deviations, follows.

```
# Pre-registration: native candidate scoring versus prompted probabilities

Committed 2026-10-12 on branch `agent/db-xps-02/vej-paper-20261012`, before
any baseline model has been run on validation or test data. Everything below
is fixed before results exist. Deviations are allowed only if recorded in the
"Deviations" section with a reason, and the paper reports them.

## Evaluated system and test

- **Vej alpha2**: Qwen3.5-2B (`15852e8c16360a2fea060d615a32b45270f8a8fc`),
  frozen BF16 backbone, rank-16 LoRA, scalar candidate head. Weights SHA-256
  `b99188eedcbc6e1faab966ed2c827025b29360157298d0cc86e53a103d73bf96`.
  Its test predictions already exist (experiment 040, PyTorch reference path)
  and are not rerun for the primary analysis.
- **Test**: r5 test, 14,286 decisions, SHA-256
  `ac1b11d62e55c6094101f71790f5bcd326b1f32afb91311bb334238036311265`.
  6,197 Choice (3-10 options), 3,089 Noul, 5,000 five-level Score.
  The test has been used once for Vej's final evaluation and once for a hosted
  comparator. Running additional baselines on it selects no Vej model.

## Baselines

| ID | System | Mode |
| --- | --- | --- |
| C-uni | Uniform distribution | control |
| C-prior | Training-split prior: per schema-family suffix (e.g. `categorical-v1`),
↪  add-one-smoothed frequency of each target key, restricted to the offered candidates
↪  and renormalised | control |
| A1 | Vej alpha1 (DeBERTa-v3-base native classifier, 384-token row limit) | native |
| Q2-L | `Qwen/Qwen3.5-2B`, same revision as Vej's backbone, untrained | answer-letter
↪  log-probabilities |
| Q2-V | same | verbalised JSON probabilities |
| Q4-L | `Qwen/Qwen3.5-4B`, BF16 | answer-letter log-probabilities |
| Q4-V | same | verbalised JSON probabilities |
| Q9-L / Q9-V | `Qwen/Qwen3.5-9B`, vLLM on-the-fly FP8 | optional; reported only if it
↪  loads and completes on the same GPU |

All prompted baselines run with thinking disabled and greedy decoding.

### Prompting protocol

- One template per mode, shared by every model. It shows the state, the
  question and every candidate's name and description as Vej receives them
  (Noul: criteria, statement, true and false descriptions; Score: each level's
  description). Candidates are labelled with capital letters in original order.
- **Letter mode**: one forward pass; the distribution is the softmax of the
  next-token log-probabilities restricted to the offered letters.
- **Verbalised mode**: the model writes a JSON object mapping each letter to a
  probability. JSON-schema guided decoding is used so keys and number syntax
  are well-formed. A response is **valid** only if every offered letter is
```

  present, every value is finite and in [0, 1], and |sum − 1| ≤ 1e-6.
- **Template tuning** uses only a fixed 300-decision validation sample
  (75 SGD categorical Choice, 75 SGD intent Choice, 75 SGD required-input Noul,
  75 HelpSteer1 helpfulness Score; seed 20261012, drawn from
  `r5/validation.jsonl`, SHA-256
  `488de164255ee1e22334a844051b1e837d5f8afa5cc7792159c93a7838ad53bc`).
  At most three variants per mode; the variant with the highest mean
  validation accuracy across Q2 and Q4 is frozen before any test row is run.
  All variants and their validation scores go in the appendix.
- **No truncation.** A prompt or candidate row longer than a model's limit is
  a failure.

Failure handling

A failure (invalid verbalised map, overlong input, runtime error) counts as
wrong for accuracy and receives the uniform distribution for NLL and Brier.
Results are also reported on valid rows only, with the validity rate.
No probability map is repaired in the primary analysis. As a separately
labelled **sensitivity analysis**, verbalised maps that are otherwise valid
but have 0 < sum and |sum − 1| ≤ 0.05 are divided by their sum.

Primary hypotheses and metrics

Primary metrics, on all 14,286 decisions, uncalibrated (temperature 1):
exact accuracy (argmax, first-candidate tie rule) and **mean NLL**
(probability floor 1e-15).

- **H1**: Vej's accuracy is higher than each of Q2-L, Q2-V, Q4-L, Q4-V.
- **H2**: Vej's NLL is lower than each of Q2-L, Q2-V, Q4-L, Q4-V.

Inference: paired differences (Vej − baseline) with a percentile bootstrap
that resamples whole `group_id`s (10,000 replicates, seed 20261012). With
eight confirmatory comparisons, a hypothesis is supported for a baseline when
the Bonferroni-adjusted 99.375% interval excludes zero in Vej's favour; it is
refuted when the interval excludes zero against Vej; otherwise inconclusive.
95% intervals are also shown. Every outcome is reported.

Secondary and descriptive analyses

None of these is confirmatory.

- Per question type, candidate count, schema family and input-length tertile.
- Summed Brier, 10-bin top-label ECE, reliability diagrams, risk-coverage.
- Score as ordinal: expected-position MAE, RPS, exact and within-one accuracy.
- Temperature: Vej with T = 1.3335 (fitted earlier on 2,026 held-out validation
  decisions, `docs/validation/alpha2-temperature.md`); letter-mode baselines
  with T fitted on the 300-decision tuning sample by the same 81-point grid.
- A1 and Q9 against Vej, and controls.
- Position bias: Q2-L and Q4-L rerun with reversed candidate order on the
  1,000-decision benchmark subset; report accuracy change and agreement.
- Vej served by vLLM on all 14,286 decisions: accuracy, NLL, top-1 agreement
  and maximum probability difference against the reference path.

Runtime benchmark (descriptive)

- Hardware and stack: one consumer GPU and one pinned vLLM image for every system.
- Workload: 1,000 test decisions, stratified by question type and input-length tertile, seed 20261012; manifest hashed before the first timed run.
- Systems: Vej (prefix caching off, then on), Q2-L, Q2-V, Q4-L, Q4-V, and Q9 if available.
- Metrics: decisions/s with all decisions submitted concurrently; p50/p95 per-decision latency at concurrency 1 after 20 warm-up decisions; server cold start to first answer; model-weight memory and configured cache budget reported from vLLM, not as a minimal memory requirement; input and output tokens. Three repeats, each with a fresh server; mean and range reported.
- Run when that machine is otherwise idle. No speed claim is made that is not in these measurements.

Deviations

Recorded during execution; none changes a primary metric or hypothesis.

1. **Prompted context length.** The protocol fixed no number; runners were set to 8,192 tokens before any run so that no test prompt fails on length.
2. **Q9 memory budget.** With the shared 85% GPU memory budget, Qwen3.5-9B FP8 loaded (10.8 GiB) but left no KV-cache memory and failed to start. Q9-L and Q9-V were rerun with a 93% budget. The budget changes memory allocation, not the computation. To keep one budget for every timed system, Q9 is excluded from the runtime benchmark.
3. **Analysis reader fix.** The shared JSONL reader split lines on Unicode line separators that occur inside JSON strings of `train.jsonl` (used only for the training-prior control). It now splits on `\n` only. The runner code that produced the predictions was not changed (commit `6bccd80`).
4. **Benchmark phases (review correction, before any timed run).** The first benchmark design measured serial latency and then concurrent throughput on the same server, so throughput would have replayed decisions already in the prefix cache. Each phase now runs on its own fresh server: serial latency over decisions 21--1,000 after 20 warm-up decisions, and throughput over decisions 2--1,000 released together after one first answer. The workloads were tokenised with the frozen code before this change.
5. **Verbalised sensitivity sums (review correction).** The sensitivity rule compared binary floating-point sums with the 0.05 bound, excluding some maps whose decimal sum is exactly on the boundary. Sums are now compared exactly in decimal, recomputed from the saved response text. Strict primary validity is unchanged.
6. **Run order.** Q9-V ran after Q9-L and before the runtime benchmark; the benchmark started only afterwards, inside the night window.
7. **Post-hoc checks requested by independent reviewers.** Vej served through vLLM was rerun (a) on the 433 Choice decisions of the benchmark sample with reversed option order and (b) on the full test with prefix caching enabled. Both are descriptive and were added after all registered results existed.